

Pablo J. Diego Simón

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🌐 <https://pjdisi.github.io/>

in pjds



Education

- Oct 2023 – Present **Ph.D. Deep Learning and Computational Neuroscience**
AI4theSciences, Marie Skłodowska-Curie Actions (MSCA)
Département d'Études Cognitives (DEC)
École Normale Supérieure (ENS), Paris

Thesis: *Linking Brain Dynamics and Large Language Models*
- Sep 2021 – Sep 2023 **M.Sc. Biomedical Engineering**
Information Technology Department (D-ITET)
ETH Zürich
Courses and scores
Thesis: *Trainability and Memory Capacity in Recurrent Neural Networks.*
- Sep 2017 – July 2021 **B.Sc. Biomedical Engineering**
Information Technology Department (ETSIT)
Universidad Rey Juan Carlos, Madrid
Courses and scores
Thesis: *A Bayesian Machine Scientist for Complex Systems' Symbolic Regression.*

Research Publications

Conference Proceedings

- 1 Pablo J. Diego-Simón and E. Niklasson, "Learning local universes: Neural Cellular Automata as decentralized world models," in *Paradigms of Intelligence, Google Research*, 2026.
- 2 P. Orhan, Pablo J. Diego-Simón, E. Chemla, Y. Boubenec, Y. Lakretz, and J.-R. King, "The Emergence of Phonemic, Syntactic, and Semantic Structures in Neural Networks: A Unified Geometrical Framework," in *Under revision.*, 2025.
- 3 Pablo J. Diego-Simón, E. Chemla, J.-R. King, and Y. Lakretz, "Probing Syntax in Large Language Models: Successes and Remaining Challenges," in *2nd Conference of Language Modeling (COLM 2025)*, 2025.
- 4 Pablo J. Diego-Simón, P. Orhan, E. Chemla, Y. Lakretz, and J.-R. King, "Polar probe decodes semantic structures from LLMs," in *3rd Conference of Language Modeling (COLM 2026)*, 2025.
- 5 Pablo J. Diego-Simón, S. D'Ascoli, E. Chemla, Y. Lakretz, and J.-R. King, "A Polar Coordinate System Represents Syntax in Language Models," in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024.
- 6 R. J. Algayres, Pablo J. Diego Simon, B. Sagot, and E. Dupoux, "XLS-r fine-tuning on noisy word boundaries for unsupervised speech segmentation into words," in *The 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. 🌐 URL: <https://openreview.net/forum?id=FGBWdf7Z19>.

Research experience

May 2026 – Sep 2026	Research Scientist Intern Paradigms of Intelligence Google Research, Zürich
Sep 2022 – Feb 2023	Research Intern Cognitive Machine Learning lab École Normale Supérieure (ENS), Paris
Jan 2022 – June 2022	Research collaborator Google Research, Zürich

Teaching

2026	(Lecturer) Advanced Data Analysis , ENS Ulm, Paris
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Awards and Certifications

Awards

2026	Selected Participant, Declaration of Digital Duties and Responsibilities , Fundación Hermes
2023	Rafael Del Pino Scholarship, From Science to Society Workshop , Christ College, University of Cambridge
2021	Award for the Outstanding Graduate of the Year , Universidad Rey Juan Carlos, Madrid. Award for the Best B.Sc. Thesis of the Year , Universidad Rey Juan Carlos, Madrid.
2020	MIT Leading Digital Transformation Scholarship (Banco Santander) , MIT PE. Excellence Scholarship , Community of Madrid, Education Delegation.
2019	Excellence Scholarship , Community of Madrid, Education Delegation. Honour Delegate Decoration , CMU Barberán y Collar.
2017	10k Fundraising for Mekong Plus Association , Institut Saint-André Ixelles.

Certifications

2020	IELTS . 8.0 Score.
2019	Test de Français International (TFI) . Maximum Score.

Skills

Technical	Python, PyTorch, JAX, CUDA, Hugging Face, Linux/Bash, Git, Slurm
Languages	Spanish — Native English — Proficient French — Proficient German — Basic